

6-DOF INDUSTRIAL ROBOT PICK-AND-PLACE

Technical Project Brief

MCTR 420 - Introduction to Robotics and Automation | Spring 2026 | Isaac Cejas and Totus Tuus Keely

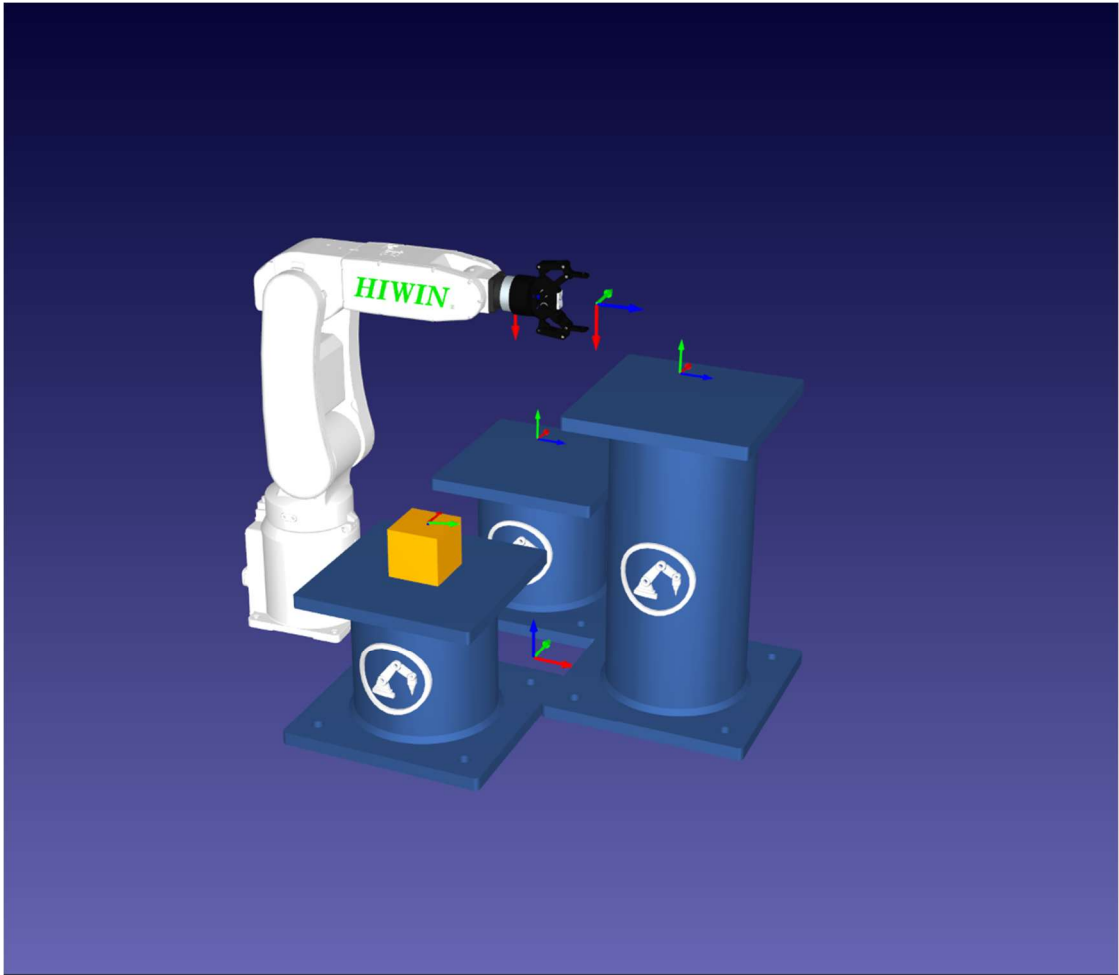
Project objective: program and validate a HIWIN RA605-710-GB robot with a RobotiQ 2F-85 gripper in RoboDK to move a box between three locations while maintaining valid poses and avoiding collisions.

Project at a Glance

Robot HIWIN RA605-710-GB	End effector RobotiQ 2F-85	Environment RoboDK	Task 3-location pick-and-place
Programming Python API	Analysis 4x4 homogeneous transforms	Validation Orthogonality + determinant	Motion Planning + collision checking

Team Roles

Primary roles: Isaac Cejas - simulation development; Totus Tuus Keely - Python programming. Both team members contributed to workcell development, motion testing, validation and documentation.



RoboDK workcell used to develop and validate the three-location pick-and-place sequence.

1. Workcell and Coordinate Frames

The simulation used a fixed robot base frame, a tool frame attached to the gripper and user-defined frames for the box, pedestals and table. Motion was referenced to the robot base so end-effector poses and transformations could be exported and checked consistently.

2. Programming and Motion Sequence

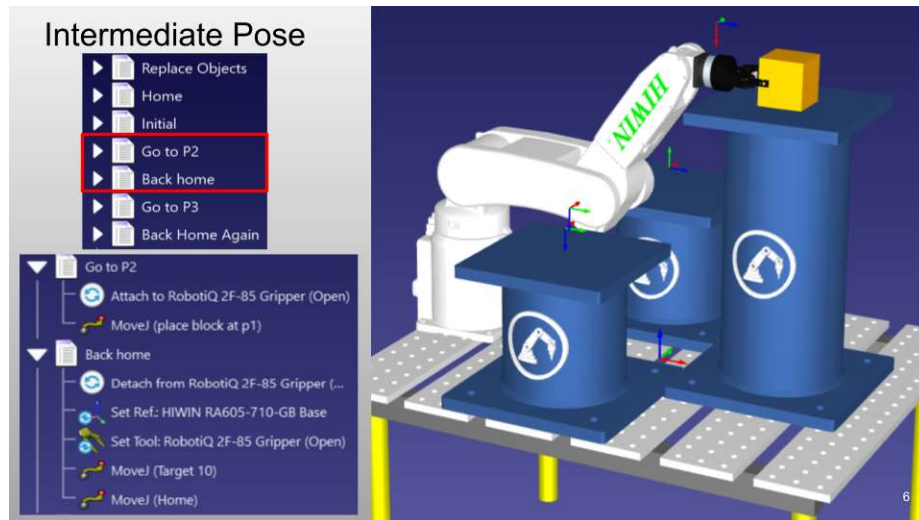
- Reset programs returned the box and robot to known starting conditions to support repeatable testing.
- Approach poses were placed above the box before pickup to reduce collision risk during attachment.
- Attach and detach events simulated gripping and releasing the box as it moved between pedestal locations.
- Python logic was integrated with the RoboDK workflow to output transformation matrices for validation.

3. Kinematic Validation

Python code output homogeneous transformation matrices for movement from home to pedestal 1, pedestal 1 to pedestal 2 and pedestal 2 to pedestal 3. The team hand-checked rotation-matrix orthogonality and the determinant condition to verify that the transformations represented valid rigid-body motion.

4. Collision-Avoidance Revision

A direct joint-motion path sometimes caused the robot or carried object to intersect a pedestal. The motion sequence was revised by adding an intermediate target that led the robot away from the obstacle before continuing to the next location.



RoboDK program tree showing the intermediate pose used during motion development.

5. Validation Result

After motion development and path revisions, the simulated robot completed the pick-and-place task between the three pedestal locations while maintaining the required box positioning and orientation. The course report states that the final sequence completed without unnecessary collisions with the pedestals or box.

Engineering Takeaways

- Coordinate frames and homogeneous transformations provided a consistent way to describe end-effector motion across the workcell.
- Hand verification of rotation matrices provided an independent check on the transformations exported from the simulation.
- Collision checking changed the motion plan: a mathematically short path was not always a usable physical path.